SU
SINAI UNIVERSITY
KANTARA

INT 453: Digital Multimedia

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Lecture 7

Wavelet Transformation Technique

Outline

- Disadvantages of DCT
- Wavelet series
- Wavelet VS Fourier transform
- How Does Wavelet Transform Work?
- Wavelet family types:
  - Discrete Wavelet Transform (DWT)
  - Haar 1D DWT
  - 2D Wavelet Transform
  - Biorthogonal Wavelets
- Wavelet applications
  - Feature extraction
  - Detecting R-Peaks in ECG Signal

Wavelet transform

This lecture will be discussed through the following video.

https://www.youtube.com/watch?v=jnxqHcObNK4

Disadvantages of DCT

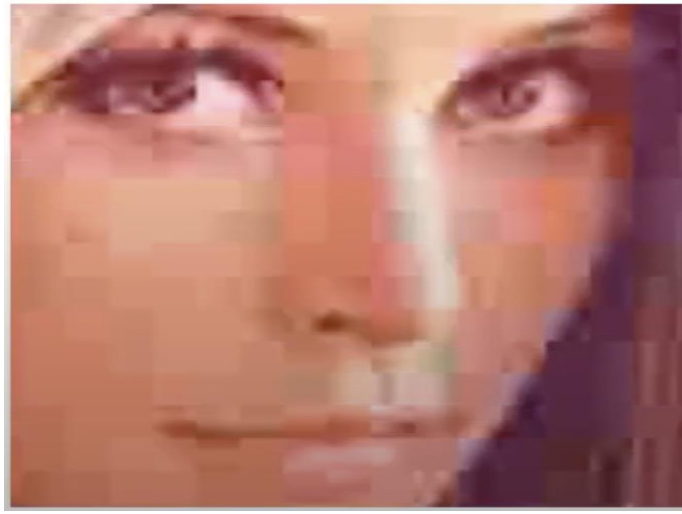
A common issue with DCT compression in digital media are blocky compression artifacts, caused by DCT blocks.

The DCT algorithm can cause block-based artifacts when heavy compression is applied.

Truncation of higher spectral coefficients results in blurring of the images, especially wherever the details are high.

Coarse quantization of some of the low spectral coefficients introduces graininess in the smooth portions of the images









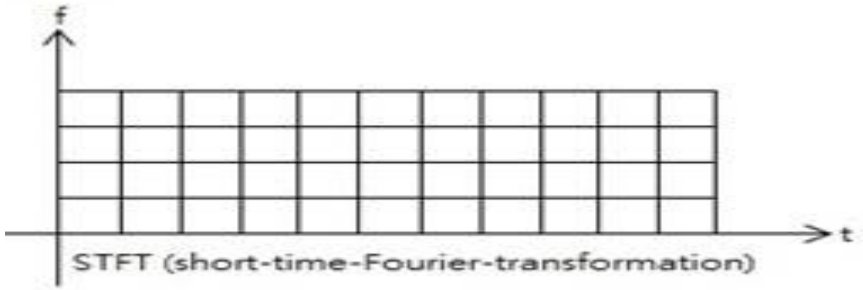


Disadvantages of DCT based JPEG

- DCT based JPEG uses blocks of image, there is still correlation across blocks.
- Block boundaries are noticeable in some cases.
- Blocking artifacts at low bit rates (there is a sharp degradation with image quality).

Wavelet series

- In mathematics, a **wavelet series** is a representation of a square-integrable (real- or complex-valued) function by a certain orthonormal series generated by a wavelet.
- wavelet-transformation contains information similar to the short-time-Fourier-transformation, but with additional special properties of the wavelets.
- Wavelet transform shows up at the resolution in time at higher analysis frequencies of the basis function.
- The difference in time resolution at ascending frequencies for the Fourier transform and the wavelet transform is shown below.





Wavelet VS Fourier transform

Transform	Representation	Input
Fourier transform	$\hat{X}(f) = \int_{-\infty}^{\infty} x(t) e^{-i2\pi ft}\, dt$	f : frequency
Time–frequency analysis	$X(t, f)$	t time; f frequency
Wavelet transform	$X(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} \overline{\Psi\left(\frac{t-b}{a}\right)}\, x(t)\, dt$	a scaling ; b time shift factor

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Wavelet VS Fourier transform

❖Fourier transforms break down signals into segments that persist over the entire sequence.	❖Wavelet transforms perform a similar function, however they can break signals down into segments localized in space and time.
❖FT decomposes signal into a set of trigonometric (sine and cosine) functions.	❖WT decomposes signal into a set of basic functions (wavelets).
❖It depends on the wave of finite duration detected in both frequency and time domains.	❖It depends on small waves detected in different times. making it as an alternative approach to eliminate the resolution problem in signal analysis.

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Figure 1



Figure 1. Representation of Fourier transform of a signal from time domain to frequency domain.

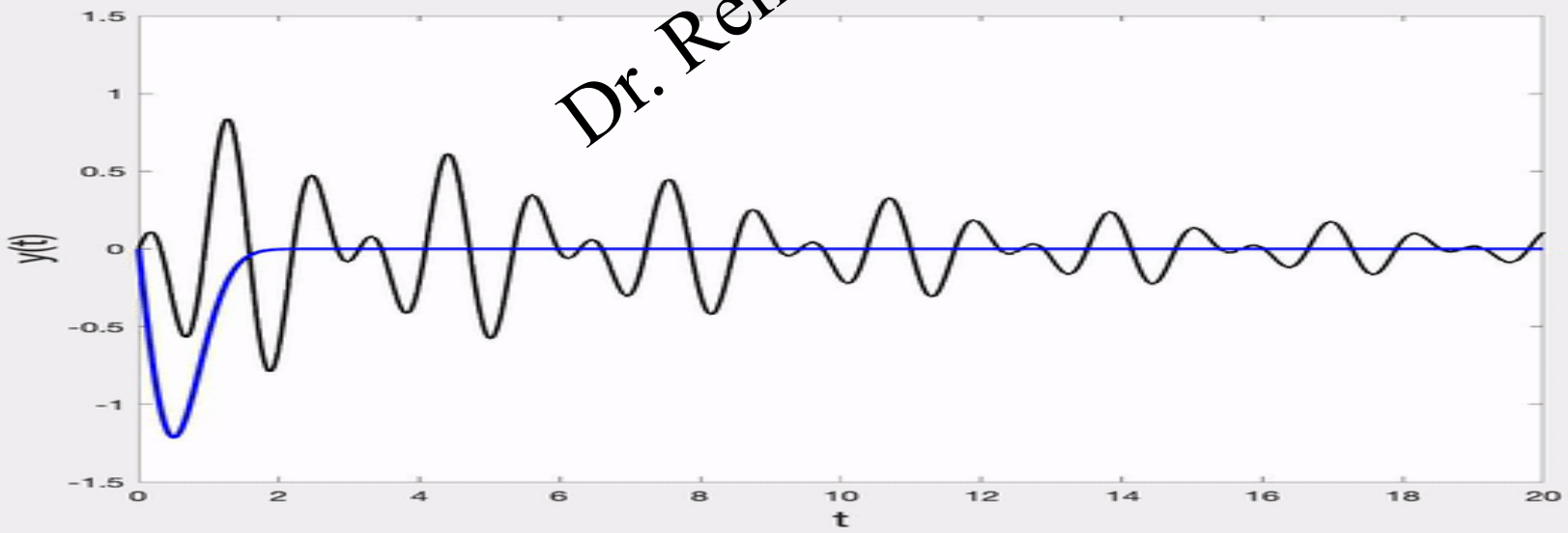
How Does Wavelet Transform Work?

- The basic idea is to compute how much of a wavelet is in a signal for a **particular scale and location**.
- For those familiar with convolutions, that's exactly what this is. A signal is convolved with a set of wavelets at a variety of scales.

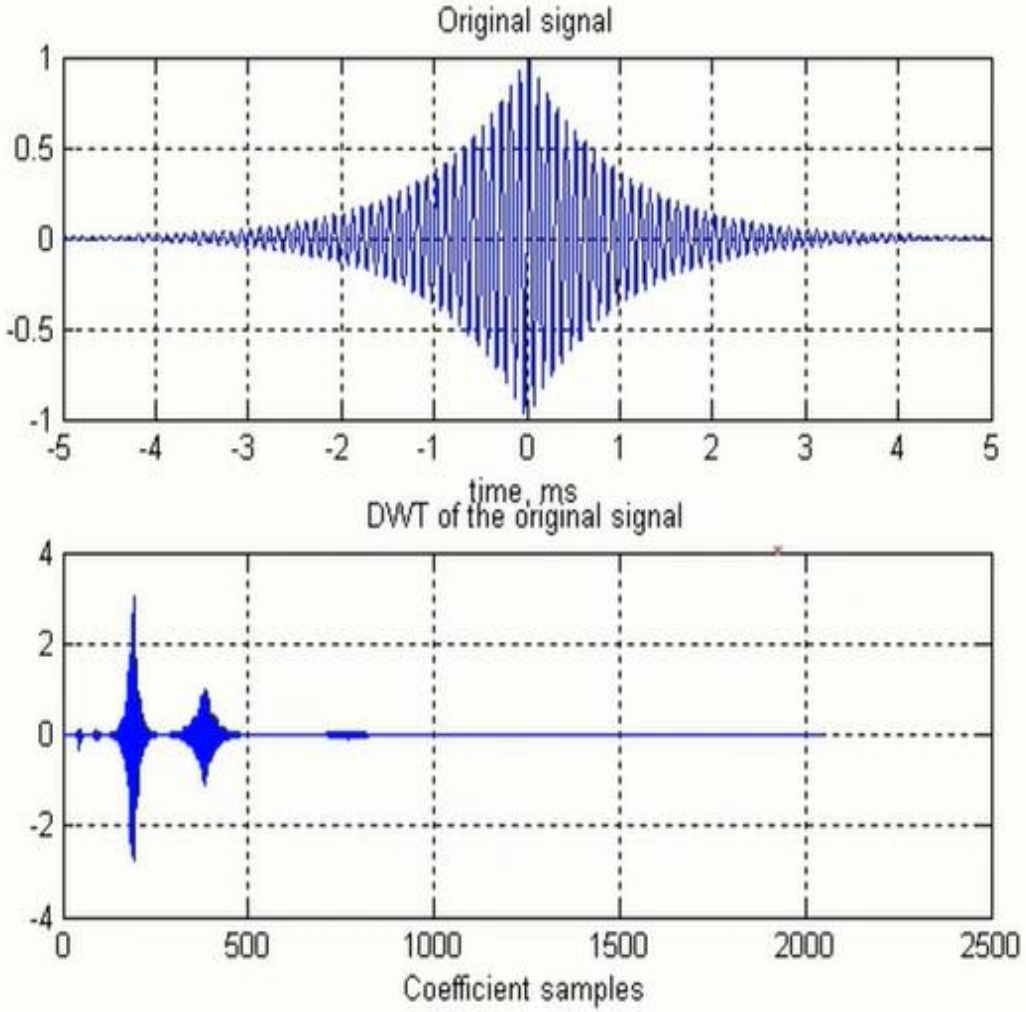
How Does Wavelet Transform Work?

- In other words, let's pick a wavelet of a particular scale (like the blue wavelet below).
- Then, slide this wavelet across the entire signal (i.e. vary its location)
- At each time step, multiply the wavelet and signal.
- The product of this multiplication gives us a coefficient for that wavelet scale at that particular time step.
- After that, increase the wavelet scale (e.g. the red and green wavelets) and repeat the process.

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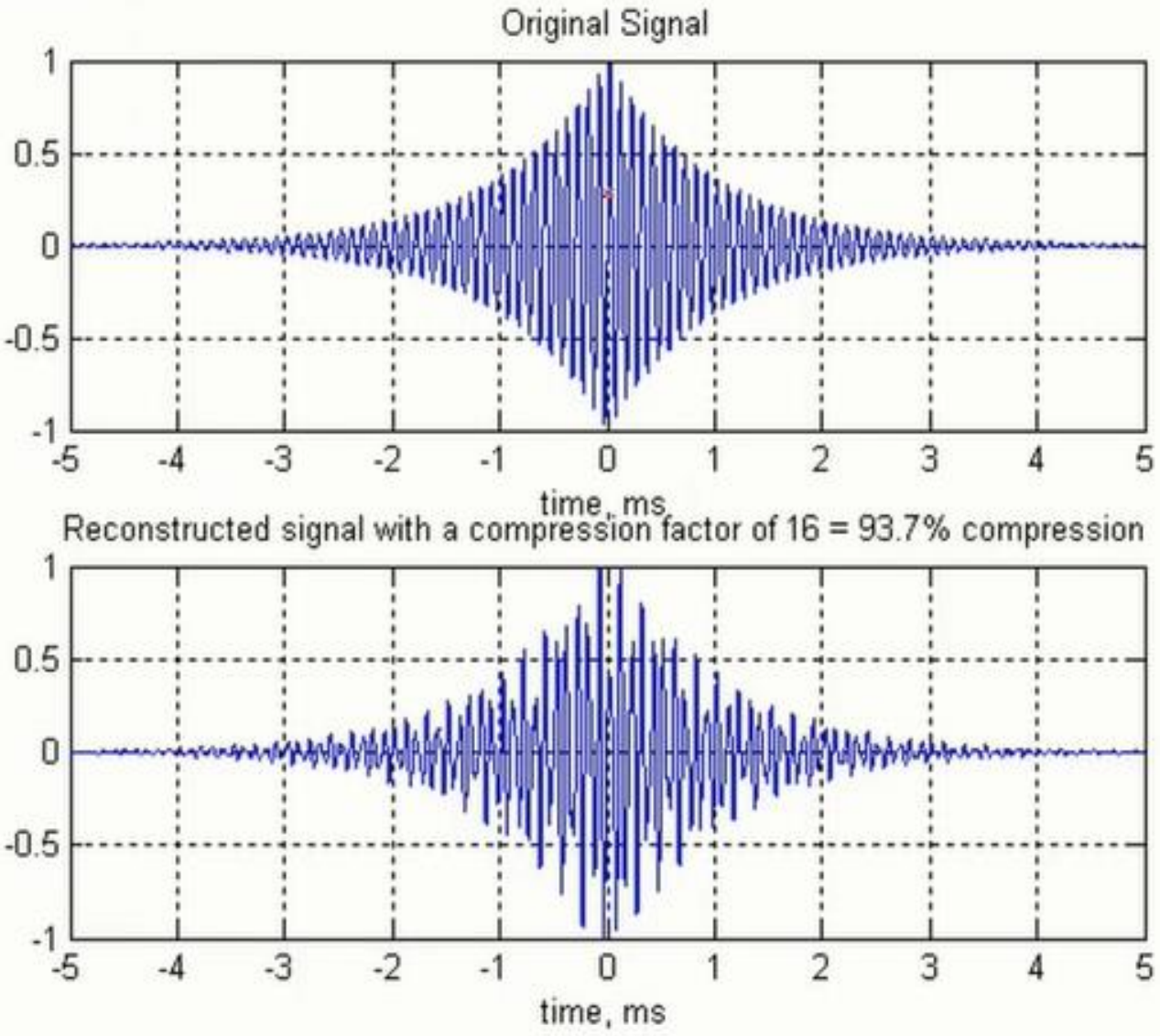


Signal Compression



DWT is commonly used for compression, since most DWT coefficients are very small, hence can be zeroed-out!

Signal Compression



Wavelet Transform

- It is the most commonly used in JPEG2000.
- The objective of the wavelet transform is to decompose the input signal into components that are easier to deal with
- It means that the input signal components have special interpretations, or
- have some components that can be **thresholded** away, for compression purposes.

Wavelet Transform (Cont'd)

- It's able to at least approximately reconstruct the original signal given these components.
- Analyzes signals in both **time and frequency** domains.
- There are two types of wavelet transforms:
  - Continuous wavelet transform (CWT) "out of scope"
  - Discrete wavelet transform (DWT)

Wavelet family types

- Orthogonal wavelets with FIR filters

  Haar, Daubechies, Symlets, Coiflets

- Biorthogonal wavelets with FIR filters

  Biorsplines

- Orthogonal wavelets without FIR filters and with scaling function

  Meyer

- Wavelets without FIR filters and scaling function

  Morlet, MexicanHat

- Complex wavelets without FIR filters and scaling function

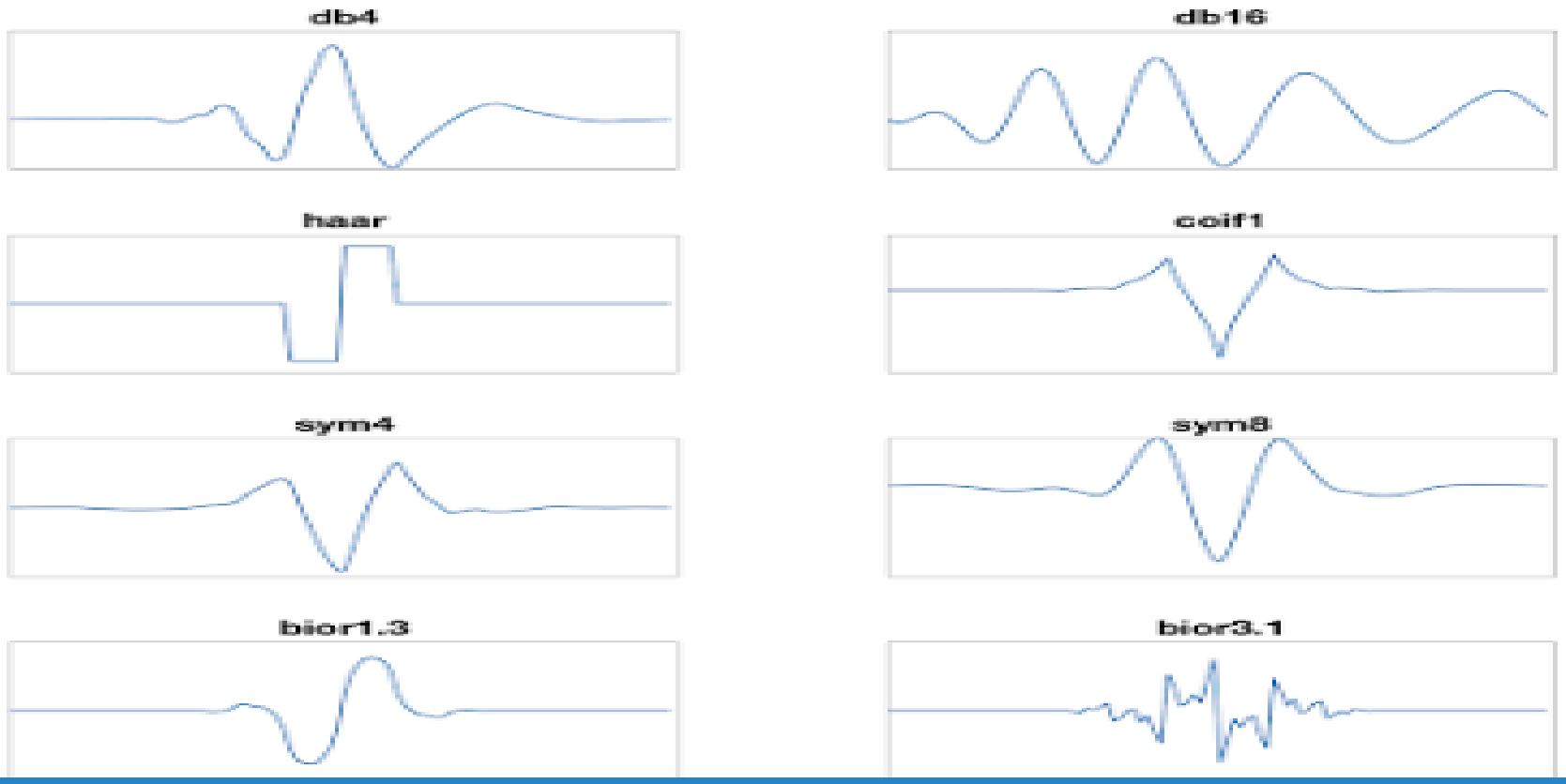
  Shannon

Various Types of wavelets

Wavelets	Abbreviations
Haar Wavelet	*Haar*
Daubechies Wavelet	*Db*
Symlets	*Sym*
Coiflets	*Coif*
Bi-Orthogonal Wavelet	*Bior*
Meyer Wavelet	*Meyr*
Discrete Meyer Wavelet	*Dmey*
Battle and Lemarié Wavelets	*Btlm*
Gaussian Wavelet	*Gaus*
Mexican Hat Wavelets	*Mexh*
Morlet Wavelet	*Morl*
Complex Gaussian Wavelets	*Cgau*
Complex Shannon Wavelets	*Shan*
Complex B-spline frequency Wavelets	*Fbsp*
Complex Morlet Wavelets	*Cmor*

Wavelet family types

If you know what characteristic shape you're trying to extract from your signal, there are a wide variety of wavelets to choose from to best match that shape.



Discrete Wavelet Transform (DWT)

- It d**ecomposes signals into scaled and shifted versions of a base function (wavelet).**
- The DWT makes the connection between wavelets in the continuous time domain
- DWT connects filters in the discrete-time domain in a multiresolution analysis framework.

Discrete Wavelet Transform (DWT)

Scaling function $$\phi(t) = \sum_{n \in \mathbf{Z}} \sqrt{2} h_0[n] \phi(2t - n)$$

Wavelet functions: $$\psi(t) = \sum_{n \in \mathbf{Z}} \sqrt{2} h_1[n] \phi(2t - n)$$

$$\psi(t) = \sum_{n \in \mathbf{Z}} (-1)^n h_0[1 - n] \phi(2t - n)$$

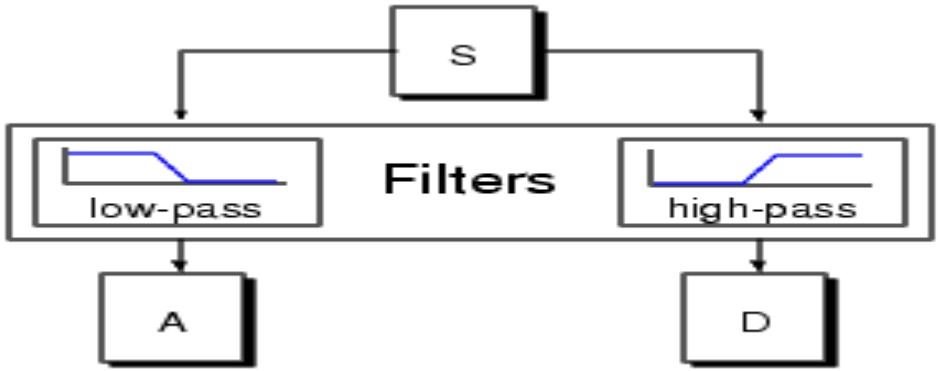
- The vectors $h_0[n]$ and $h_1[n]$ are called the low-pass and high-pass ***analysis filters***.
- To *reconstruct* the original input, an inverse operation is needed.
- The inverse filters are called ***synthesis filters***.

Discrete Wavelet Transform (DWT)

❖Decompose the signal into coarse approximation and details

❖The **first half** of the new sequence contain averages from the original sequence, so we can view it as a coarser approximation to the original signal.

❖The **second half** of new sequence can be viewed as the details or approximation errors of the first half.



Coarse approximation

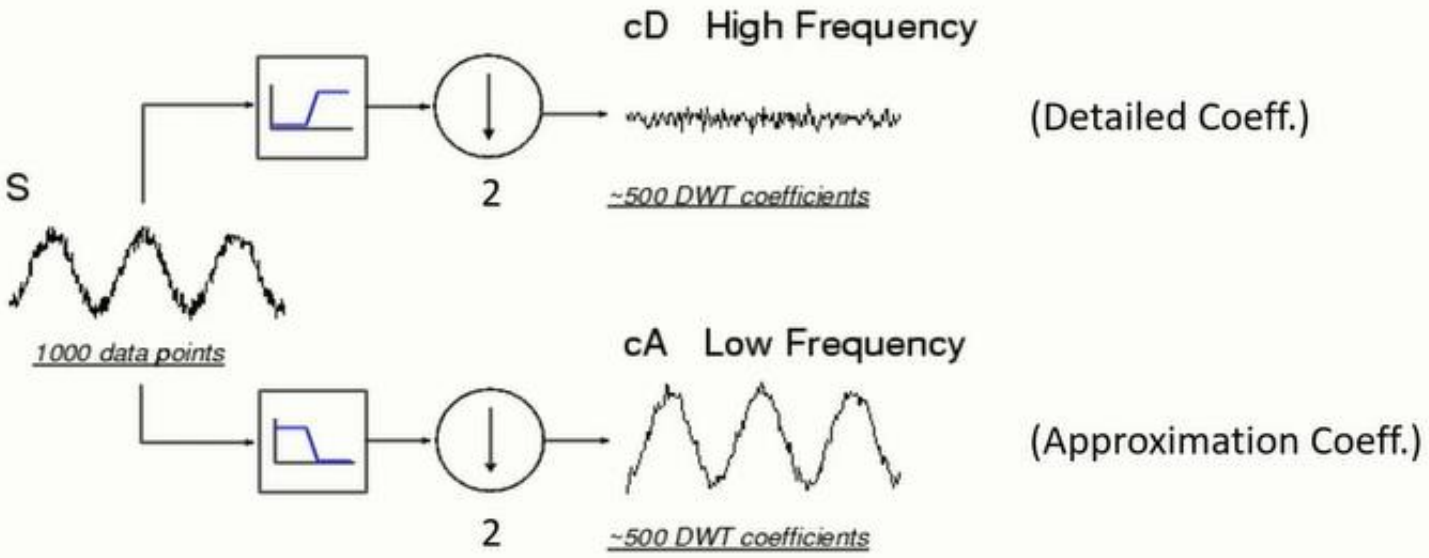
Details

DWT

The wavelet decomposition of a signal $s(t)$ based on the multi resolution theory given by **S. Mallat** and **Meyer** can done using digital FIR filters as shown in figure.

One level Decomposition

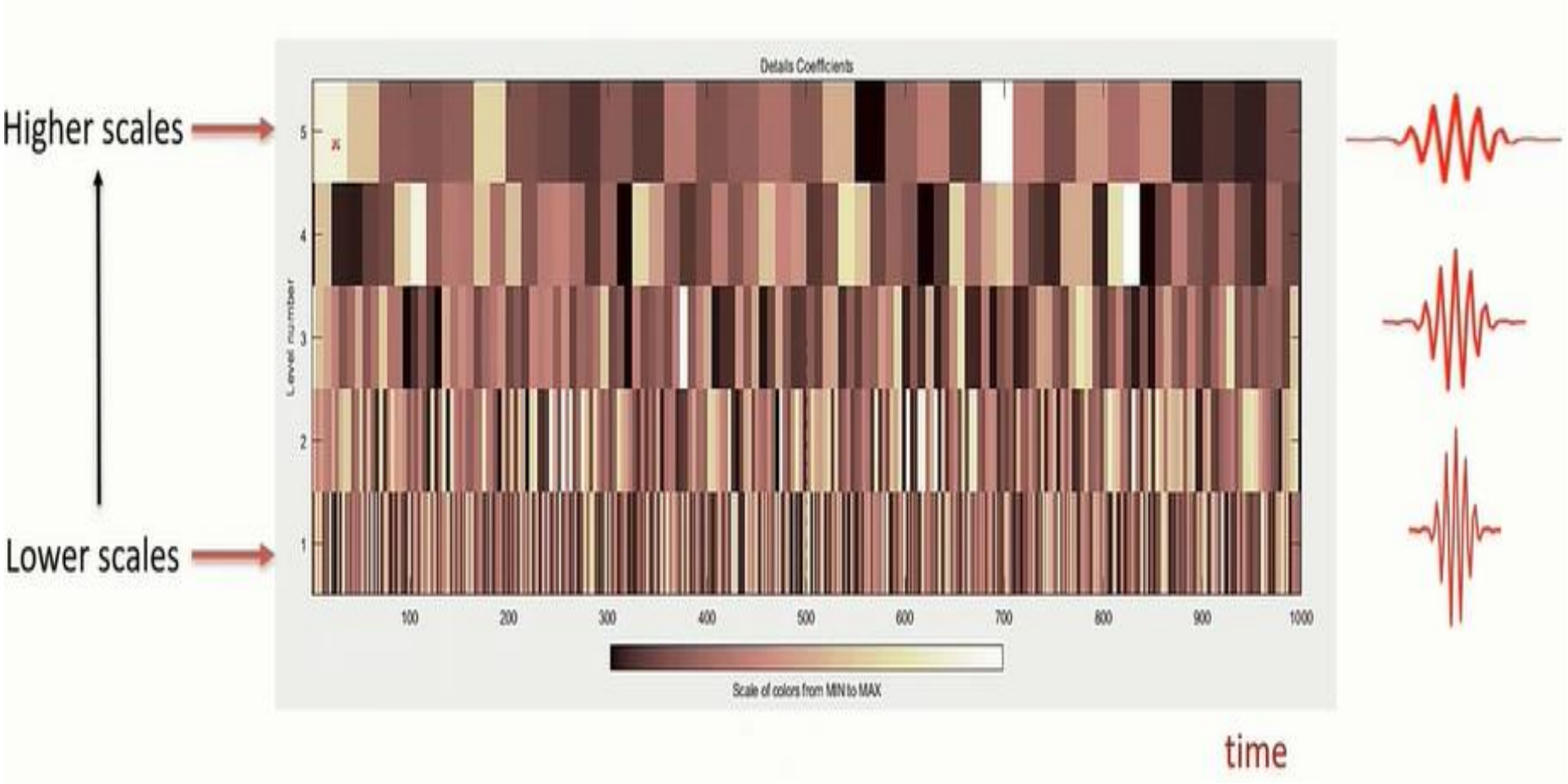
Signal Analysis



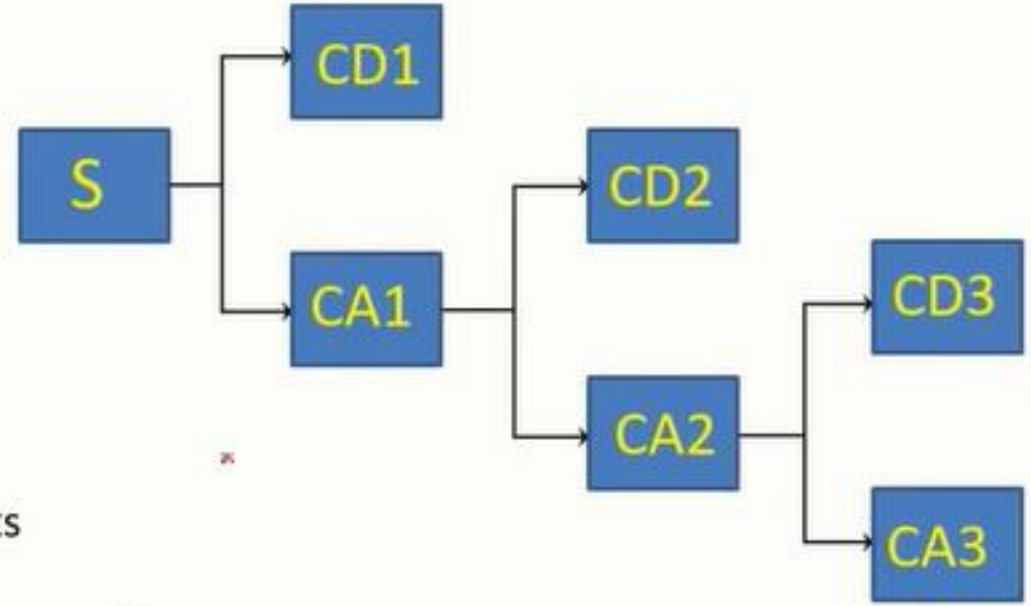
Key advantages of the wavelet transform are:

- Wavelet transform can extract local spectral and temporal information simultaneously.
- There are a variety of wavelets from which to choose.

Scalogram: DWT coefficients Representation



DWT decomposition

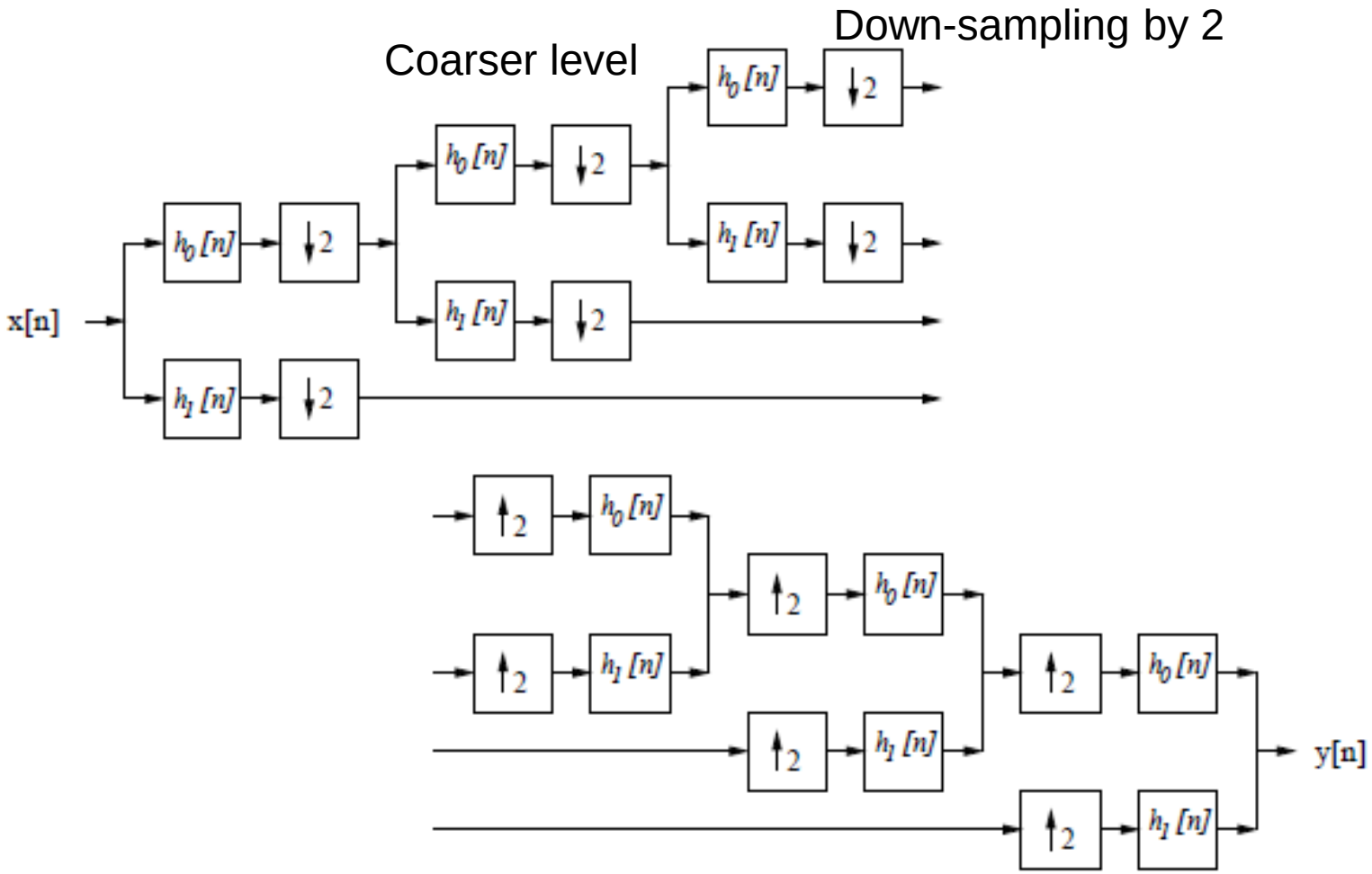


Here,
S: Signal,
CD: Detailed Coefficients
CA: Approximation Coefficients

This is called Wavelet decomposition tree.

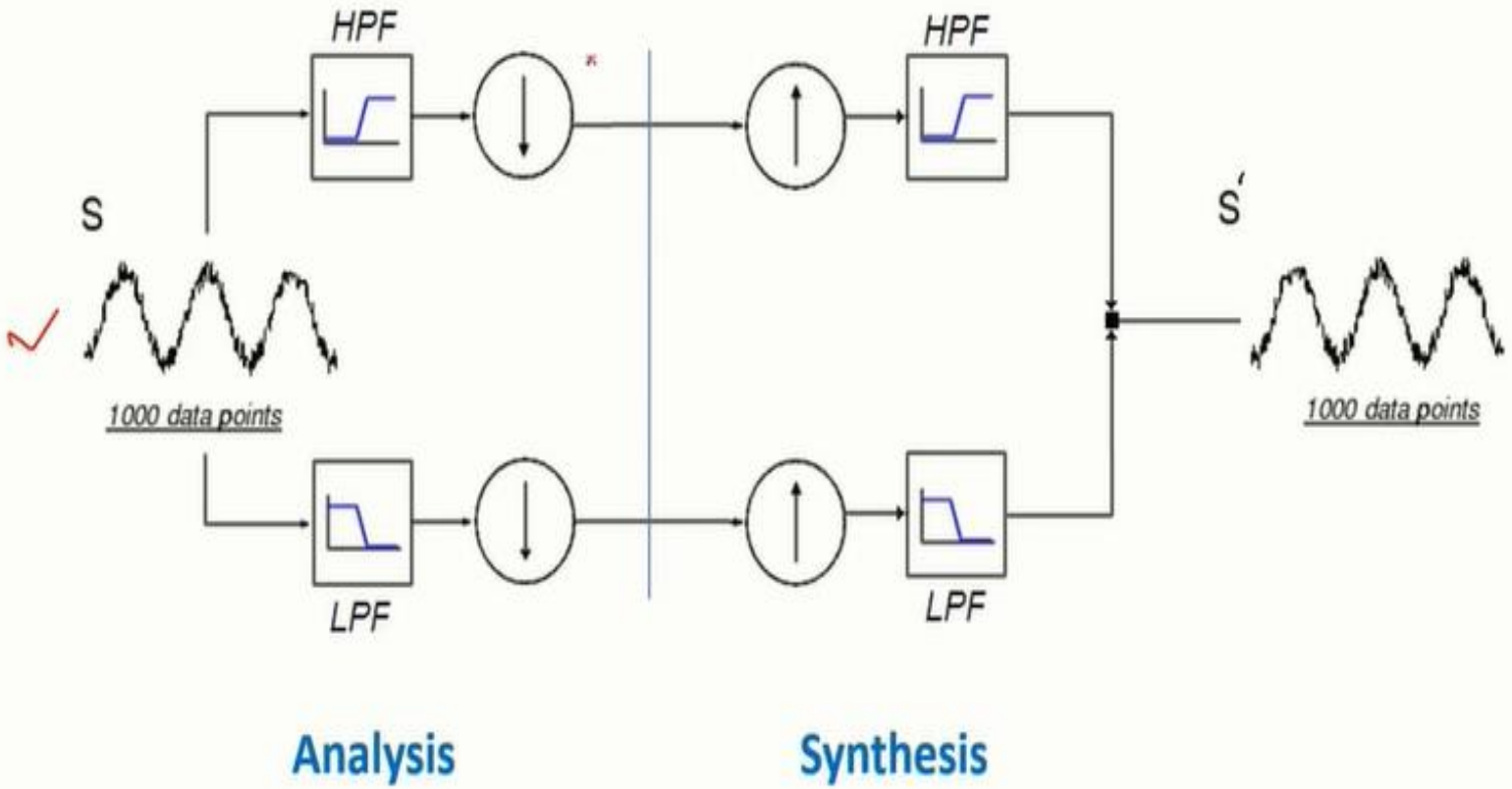
Max. Number of decomposition levels: $\log_2 N$

Block Diagram of 1D Wavelet Transform



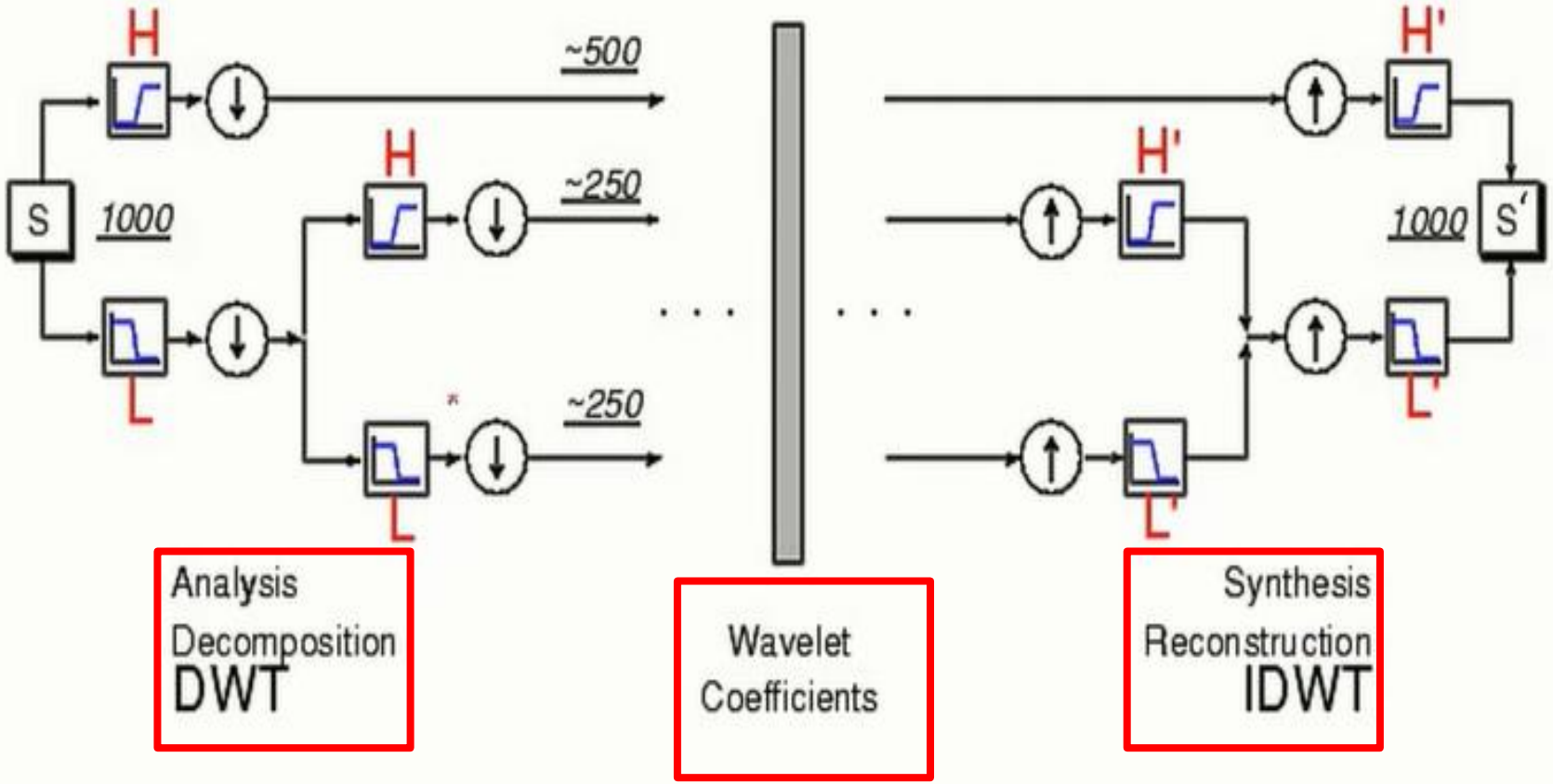
Discrete Wavelet Transform (Filter Banks)

Analysis & Synthesis



Discrete Wavelet Transform (Filter Banks)

Multistep Decomposition and Reconstruction



Discrete Wavelet Transform (Filter Banks)

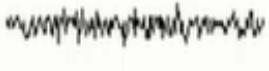
IDWT: Inverse Discrete Wavelet Transform

Signal Synthesis

One level Reconstruction

(Detailed Coeff.)

cD High Frequency



~500 DWT coefficients

HPF

2

S'

1000 data points

cA Low Frequency



~500 DWT coefficients

2

LPF

pproximation Coeff.)

Haar 1D DWT

- The number of elements in each set $\{x_{n-1,i}\}$ and $\{d_{n-1,i}\}$ is exactly **half** of the number of elements in the original sequence.
- Form a new sequence $\{x_{n-1,i}, d_{n-1,i}\}$. having length equal to that of the original sequence by concatenating the two sequences $\{x_{n-1,i}\}$ and $\{d_{n-1,i}\}$.

Haar 1D DWT

It is easily verified that the original sequence can be reconstructed from the transformed sequence using the relations

$$x_{n,2i} = x_{n-1,i} + d_{n-1,i}$$
$$x_{n,2i+1} = x_{n-1,i} - d_{n-1,i}$$

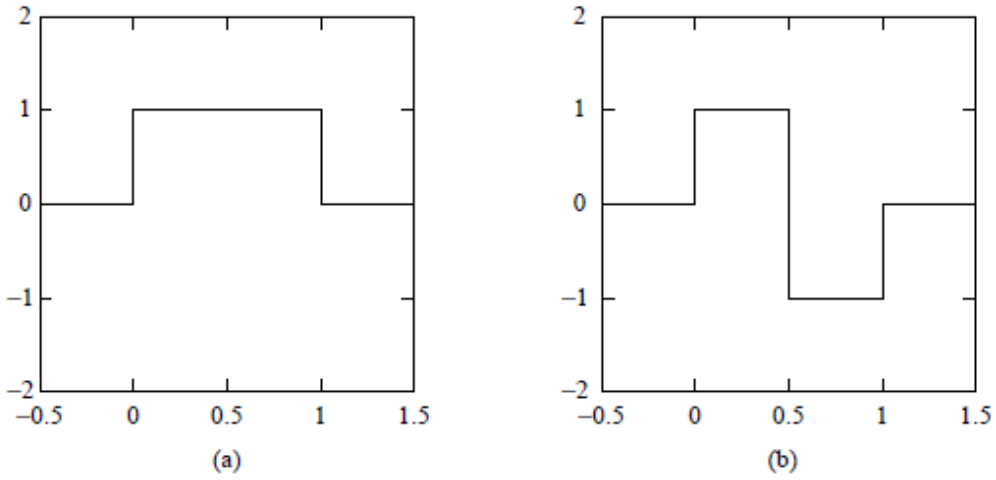


Fig. 8.12: Haar Transform: (a) scaling function, (b) wavelet function.

2D Wavelet Transform

For an NxN input image, the two-dimensional DWT proceeds as follows:

- Convolve each row of the image with $h_0[n]$ and $h_1[n]$,
- Discard the odd numbered columns of the resulting arrays,
- Concatenate them to form a transformed row.
- After all rows have been transformed, convolve each column of the result with $h_0[n]$ and $h_1[n]$.
- Again discard the odd numbered rows and concatenate the result.

2D Wavelet Transform

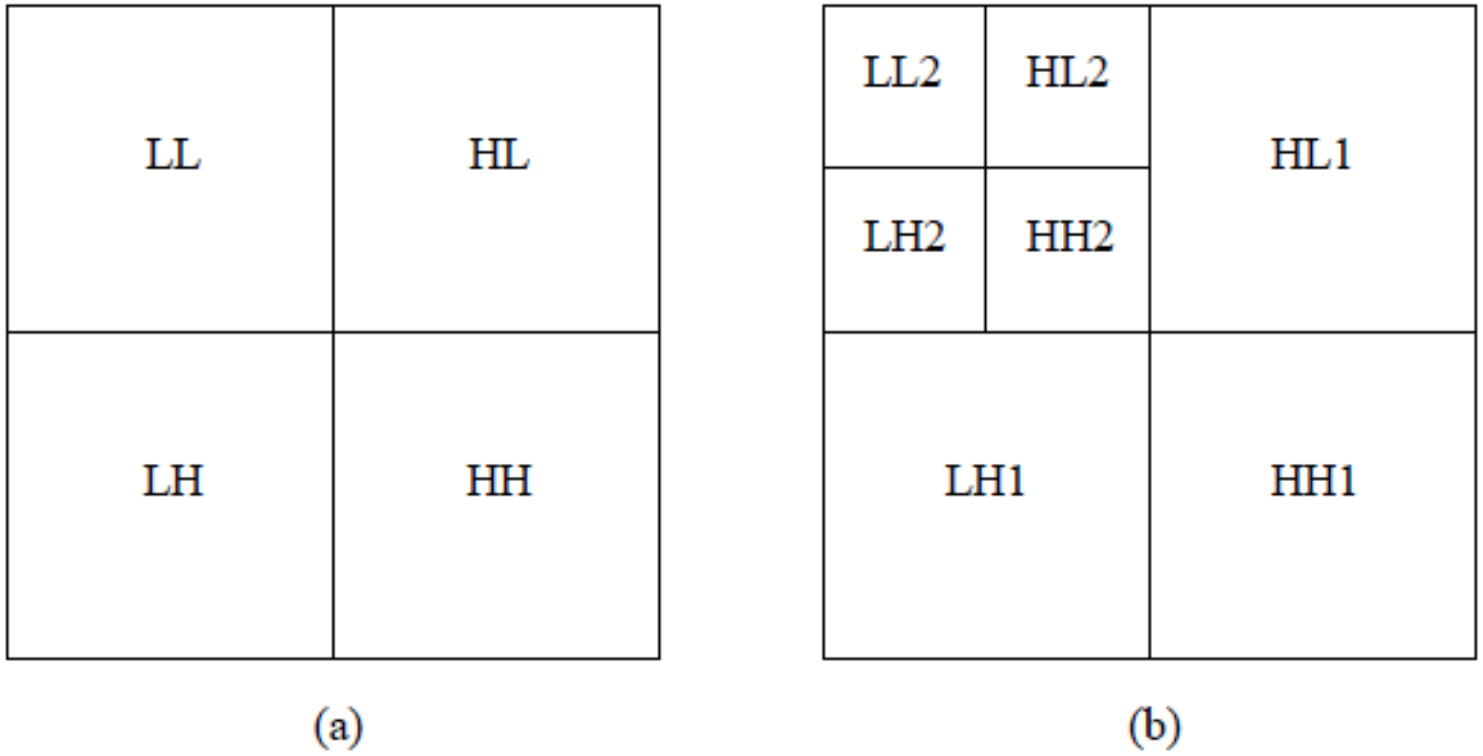
After the above two steps, one stage of the DWT is complete.

The transformed image now contains four sub-bands LL, HL, LH, and HH, standing for low-low, high-low, etc.

The LL sub-band can be further decomposed to yield yet another level of decomposition.

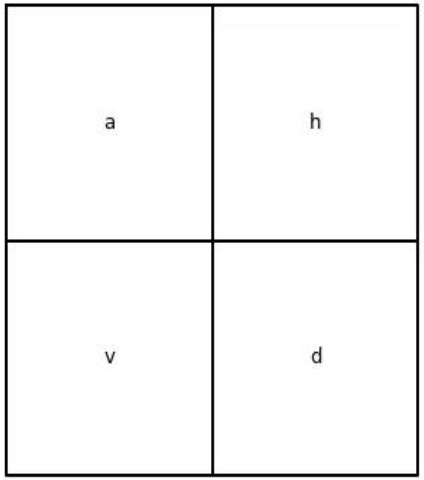
This process can be continued until **the desired number of decomposition levels** is reached.

2D Wavelet Transform

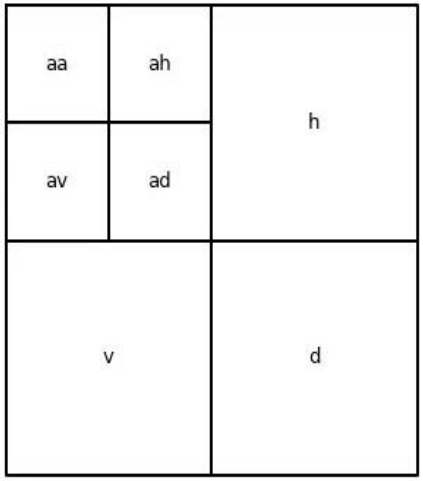


The 2DWT (a) one level transform (b) two level transform

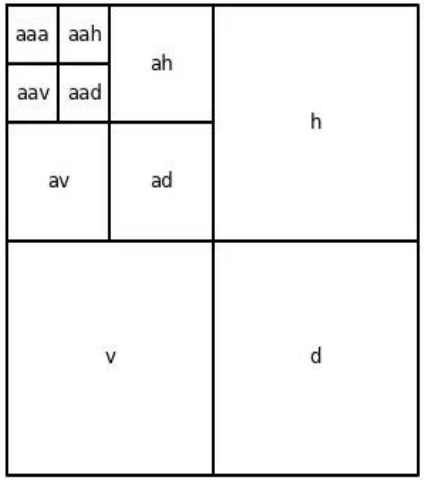
1 level decomposition



2 level decomposition



3 level decomposition



Image



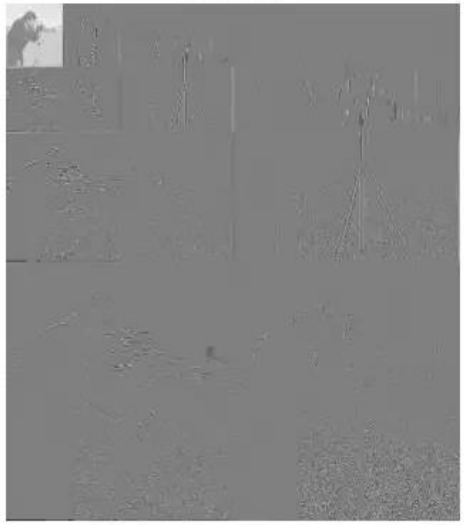
Coefficients (1 level)



Coefficients (2 level)



Coefficients (3 level)

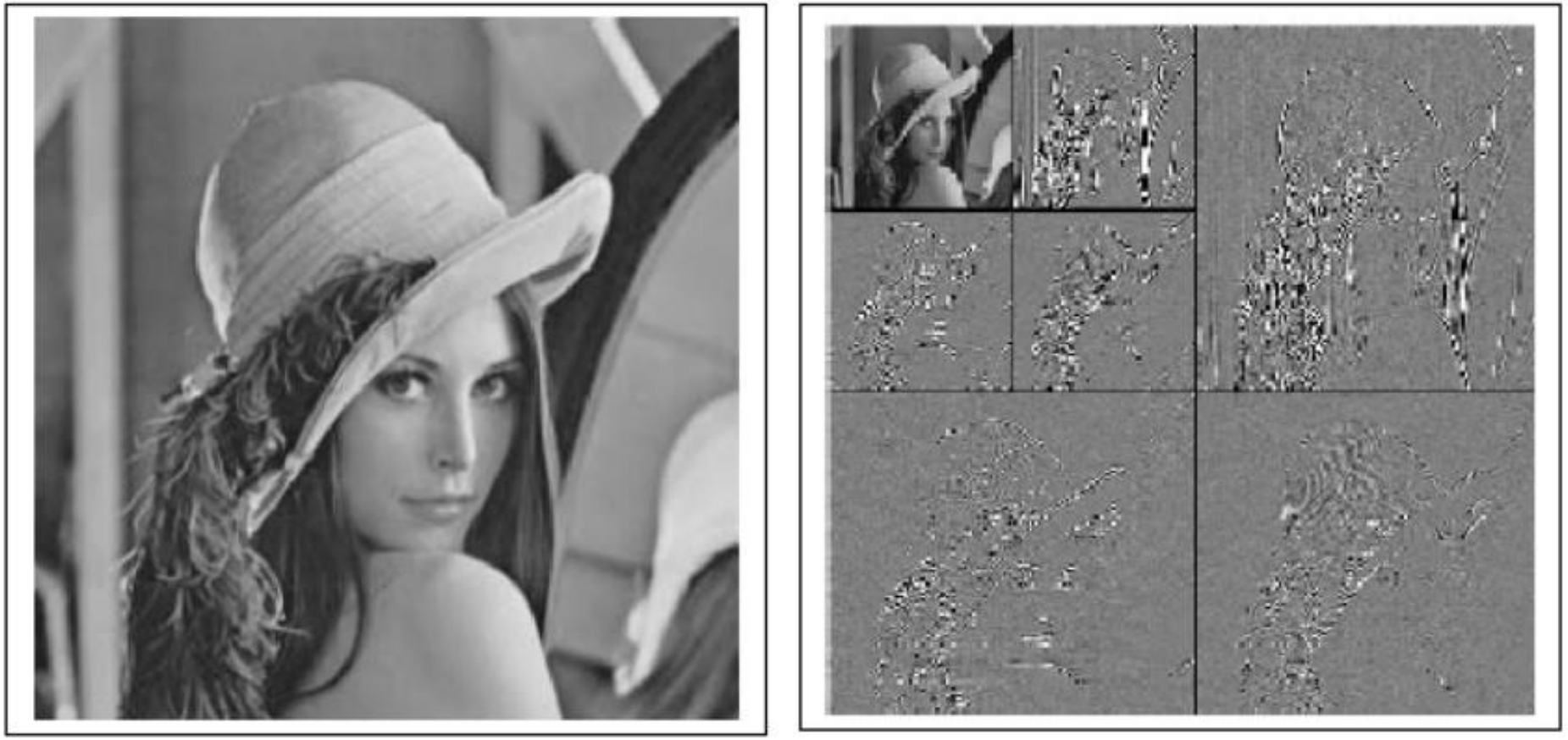


2D Wavelet Transform



shows the image Lena after one Haar wavelet transform

2D Wavelet Transform



2 level 2-D transform on a Lena image

Biorthogonal Wavelets

- For **orthonormal** wavelets, the **forward transform and its inverse are transposes of each other and**
- **The analysis filters are identical to the synthesis filters.**
- **Without orthogonality**, the wavelets for analysis and synthesis are called biorthogonal".
- **The synthesis filters are not identical to the analysis filters**
- They can be denoted as $\sim h0[n]$ and $\sim h1[n]$.
- To specify a biorthogonal wavelet transform, we require both $h0[n]$ and $\sim h0[n]$.

$$h_1[n] = (-1)^n \tilde{h}_0[1-n]$$

$$\tilde{h}_1[n] = (-1)^n h_0[1-n]$$

Various wavelet families

Wavelets with filters			Wavelets without filters	
With compact support		With non-compact support	Real	Complex
Orthogonal	Biorthogonal	Orthogonal		
Db, haar, sym, coif	*bior*	*meyr, dmey, btlm*	*gaus, mexh, morl*	*cgau, shan, fbsp, cmor*

Wavelet applications

- Physics(acoustics, astronomy, geophysics)
- Telecommunication Engineering(signal processing, subband coding, speech recognition, image processing, image analysis)
- Mechanical engineering(turbulence)
- Medical (digital radiology, computer aided diagnosis, human vision perception)
- Applied and Pure Mathematics(fractals)

Wavelet applications

Table 1: Classification of Wavelets

Area(s) of Application	Type of Wavelet
Human Vision Perception	Biorthogonal wavelets [43-44]
Identification of Events with Specific Frequencies	Shannon wavelets [45-46]
Edge Detection and Reconstruction of Binary Pulses	Haar Wavelets [47-49]
Fractals - Data with self-similarities	Coiflets Wavelets [50-53]
Signal Processing and Data Compression	Daubechies Wavelets [54-57]
Vibration and Sound	Gaussian and Morlet Wavelets [58-59]
Biomedical Signal Analysis	Mexican Hat Wavelets [60-61]

COPY

Comparative analysis

Feature	Fourier Transform	Wavelet Transform
Analysis Domain	Global (frequency)	Local (time-frequency)
Basis Functions	Sinusoids	Scaled and shifted wavelets
Applications	Stationary signals	Non-stationary signals
Computational Cost	Efficient for periodic data	Higher for detailed analysis

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Wavelets for feature extraction

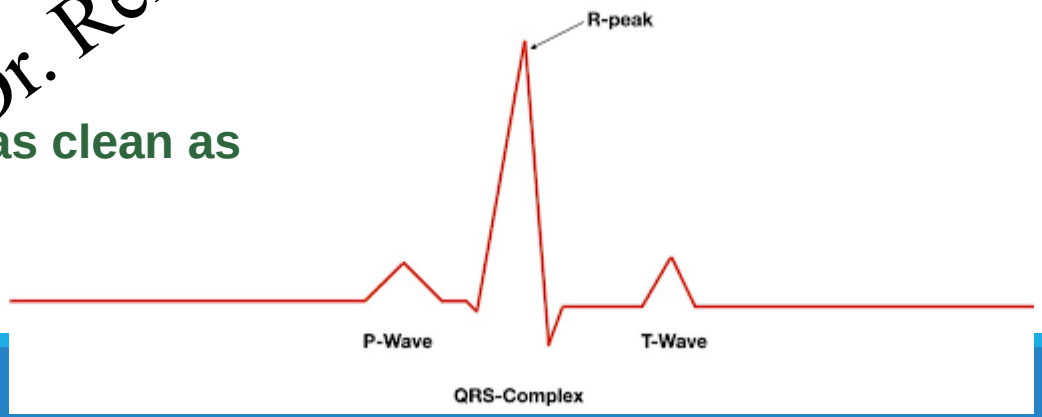
- **smaller support wavelets such as haar, db2,** or **sym2 are used to extract features.**
- The support of the wavelet should be small enough to separate the features of interest.
- Wavelets with larger support tend to have difficulty in detecting closely spaced features.
- Using wavelets with large support can result in coefficients that do not distinguish individual features.

Detecting R-Peaks in ECG Signal

- In this example, a type of discrete wavelet transform is used to detect R-peaks from an electrocardiogram (ECG).
- ECG measures heart activity.
- R-peaks are typically the highest peak in an ECG signal.
- They are part of the QRS-complex, which is a characteristic oscillation that corresponds to the contraction of the ventricles and expansion of the atria.
- Detecting R-peaks is helpful in computing heart rate and heart rate variability (HRV).

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ECG signals that look as clean as this graphic



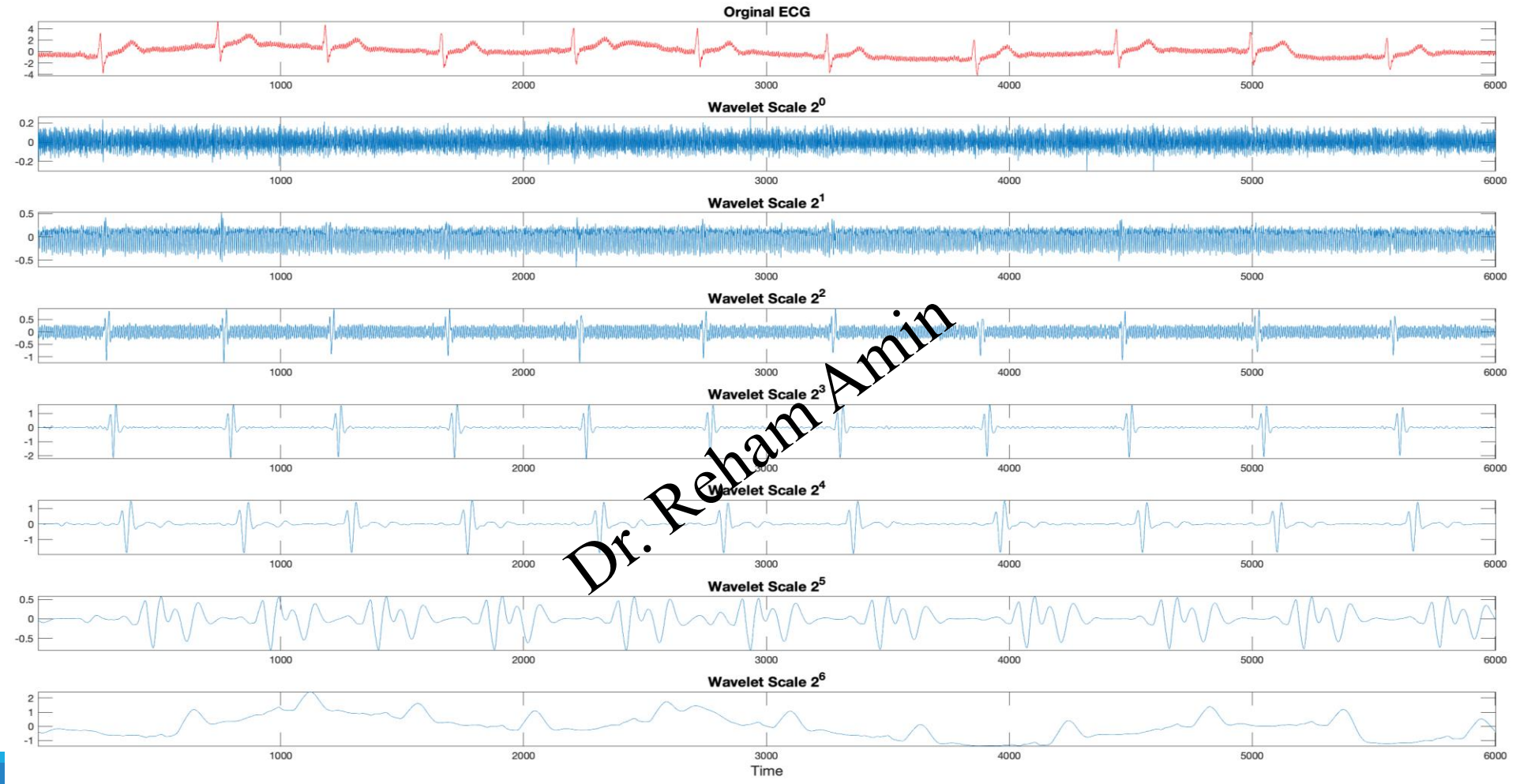
Detecting R-Peaks in ECG Signal

- ECG data is typically noisy.
- For R-peak detection, simple peak finding algorithms will fail to generalize when applied to raw data.
- The wavelet transform can help convert the signal into a form that makes it much easier for our peak finder function.
- The maximal overlap discrete wavelet transform (MODWT) is used to extract R-peaks from the ECG waveform.

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Detecting R-Peaks in ECG Signal

The original ECG signal along with wavelet coefficients for each scale over time.



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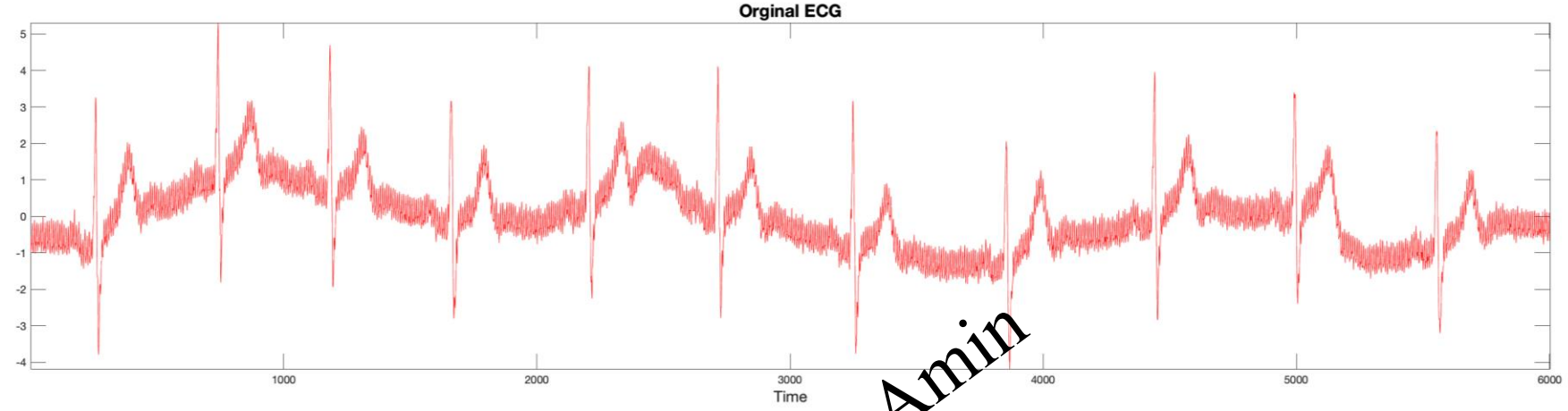
Detecting R-Peaks in ECG Signal

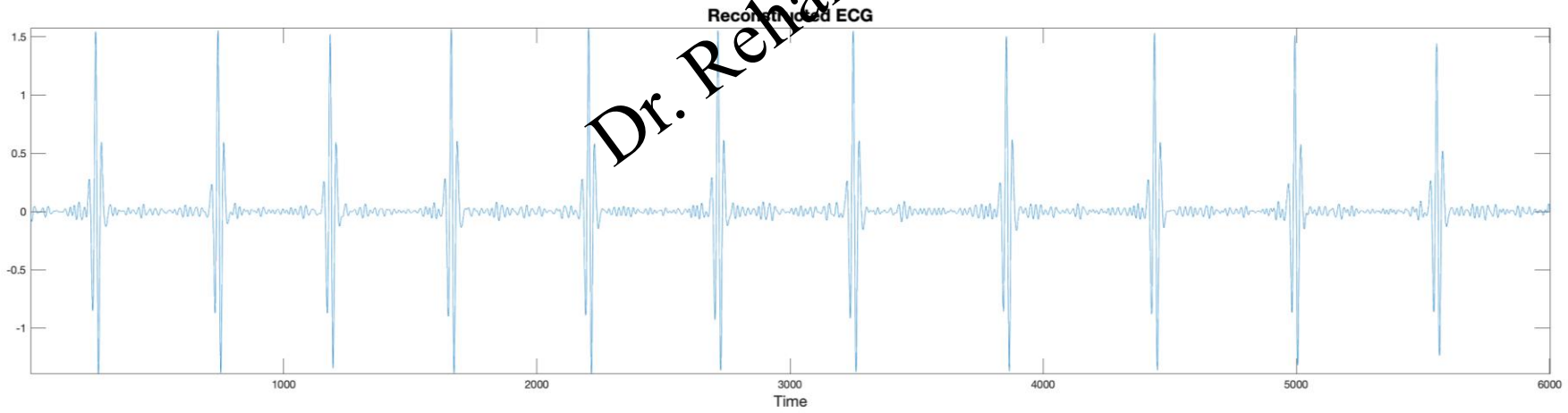
- https://builtin.com/data-science/wavelet-transform

- https://github.com/ShawhinT/YouTube-Blog/tree/main/waveletTransform

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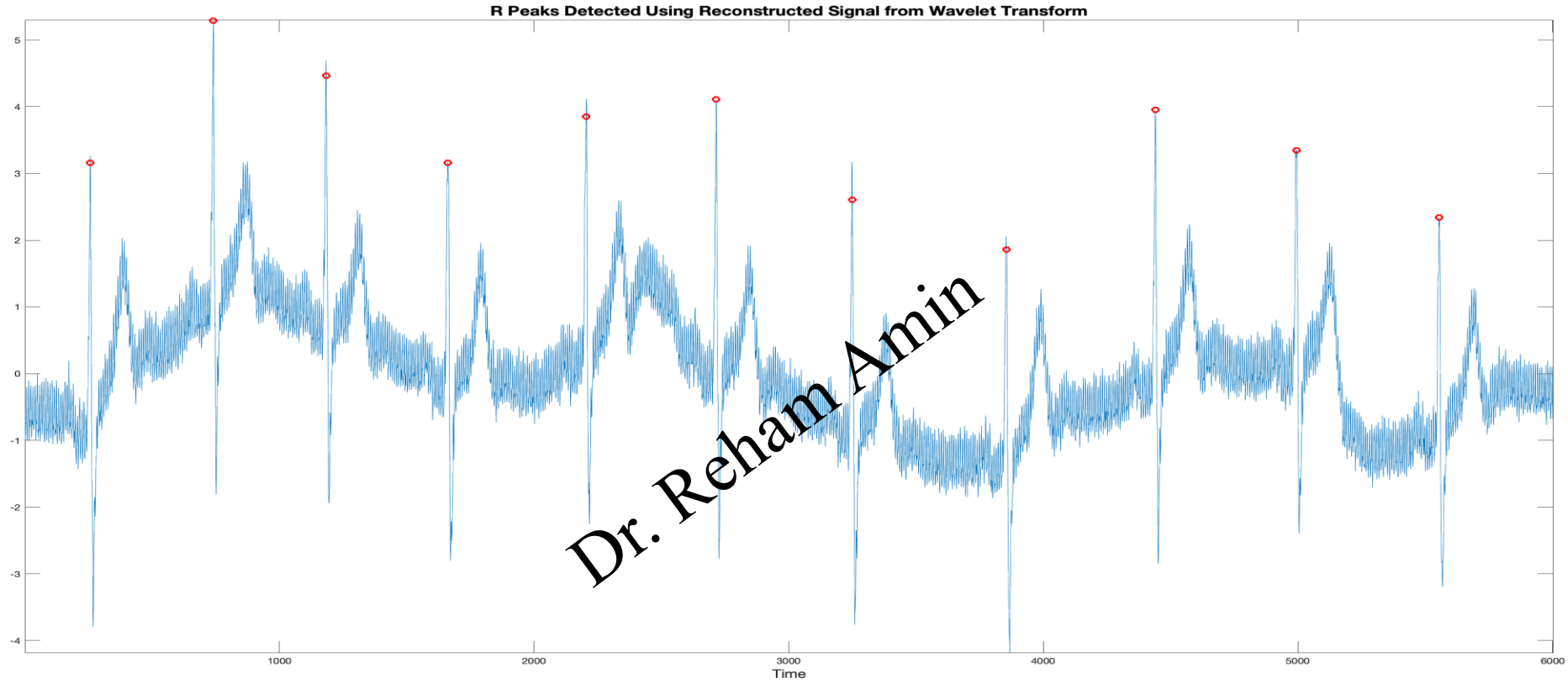
Detecting R-Peaks in ECG Signal





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Detecting R-Peaks in ECG Signal



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Assignment

- How to Choose a Right Wavelet and Wavelet Transform, explain your answer by examples?

- What is the advanced development in wavelet?

Review questions

1. What is the benefit of using wavelet transform in compression?
2. Why wavelet transform is preferred over DCT in the digital signal processing field?
3. Compare between the output of the scaling and wavelet functions in the wavelet transform.
4. Compare between orthogonal and biorthogonal wavelet.

Questions ?

Thank You